

Row 3: Bond Valuation (Chapter 6)

Scenario

"SafePort Inc." issued 25-year bonds 3 years ago. The bonds have a par value of 1,000, ~~payan~~81,000, ~~payan~~81,080. Since issuance, SafePort's credit rating has improved, and similar-risk bonds now yield 6.0% to maturity. The bond indenture also contains a sinking-fund provision that requires the firm to retire 4% of the original issue each year, starting in year 8. The retirements may be made by purchasing bonds in the open market or by calling them at par (\$1,000) via lottery.

An investor buys one bond today at the market price and intends to hold it for 5 years. She estimates that there is a 60% probability the bond will be called at the first call date; otherwise, she will hold it to maturity (year 22). For simplicity, assume that if the bond is not called at the first call date, it will not be called at all, and the sinking-fund retirements do not affect her bond (it is never selected in the lottery).

Questions

- (a) Compute the bond's yield to maturity and yield to call (assuming call at the first call date).
- (b) Calculate the expected horizon yield (total return) for the 5-year holding period under the investor's probability estimate. (Hint: If the bond is called at the end of year 4, she receives the call price plus the year-4 coupon; if not called, she sells the bond at the end of year 5 at a price consistent with the then-prevailing YTM for a bond with 17 years to maturity, assuming the YTM remains at 6.0%.)
- (c) Explain why the bond's market price today is likely above par. How does the sinking-fund provision affect the bond's risk and price? What would be the impact on the bond's value if the sinking-fund retirements were executed at the call price instead of at par?